
What Does a Benign-Trained Anomaly Detector Actually Detect?

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Abstract

Anomaly detectors for insider threats are usually judged by how well their scores rank malicious activity above normal activity. We show that this number can be misleading, because it does not reveal what information the detector uses. We build a detector by adapting a large language model to normal employee activity only, so that unusual activity receives a high surprisal score, and we then open the model to determine what its score is based on. On one standard benchmark the evidence the audit selects turns out to be almost entirely a description of who each user is (their recorded personality and organizational profile) rather than of what they did; on a second benchmark it is mostly behavior. When malicious users are compared only against normal users that the model has never seen, apparent ranking quality falls from about 0.95 to near chance on the first benchmark. The same analysis on a third dataset, collected from real people with real personality questionnaires, confirms the behavioral case, and classical statistical detectors trained on the same information do not take the profile shortcut. Retraining the detector on text with the profile removed shows that, for the large model we audited, the profile is the cause and not a bystander: removal restores ranking of unseen users from chance to about 0.90; for a smaller model the same removal makes little measurable difference, a contrast between two particular models that our design does not attribute to size alone. On a corporate login log with no profile of any kind, the same failure appears and the shortcut moves to the computer names, which are associated with users without being fixed per user. The conclusion is that high ranking performance alone does not show that a detector has learned to detect behavior, and we provide a way to check.

1 The problem

Organizations record what employees do on their computers: logons, file transfers, web activity, e-mail volume, and so on. An *insider threat* is an employee who misuses legitimate access, for example by stealing data. Because such events are rare, a common approach is *anomaly detection*: build a statistical model of normal activity, and flag days that the model finds unusual.

Formally, a detector is a function s that assigns a real-valued *anomaly score* $s(x)$ to each observation x , where in our setting x is a record of one user’s activity on one day. Higher scores are meant to indicate more suspicious activity. Detectors are usually evaluated by how well their scores *rank* malicious days above benign ones. The standard summary of ranking quality is the following probability, estimated over a labeled test set:

$$\text{AUC} = \Pr(s(X_{\text{mal}}) > s(X_{\text{ben}})) + \frac{1}{2} \Pr(s(X_{\text{mal}}) = s(X_{\text{ben}})), \quad (1)$$

where the second term counts tied scores as half a success, X_{mal} is a randomly chosen malicious day and X_{ben} a randomly chosen benign day. (This quantity is commonly called the ROC-AUC, for “area under the receiver operating characteristic curve”; the probability formulation above is equivalent and is the only form we need.) An AUC of 1 means perfect ranking and 0.5 means the ranking is no better than chance.

The question that motivates our work is simple to state. A high AUC says that the score *separates* the two classes. It does not say *on the basis of what information* the separation is achieved. Two detectors with identical AUC could work very differently: one might respond to genuinely unusual behavior, while the other might respond to incidental properties of *who* the malicious users happen to be, such as their role, department, or personality profile. The second kind of detector would be nearly useless in practice, because in deployment the malicious users are not known in advance. Our paper develops a method for determining which kind of detector one actually has, and applies it to a modern detector built from a large language model.

2 The data

We use two versions (called r4.2 and r6.2) of the CERT insider-threat corpus, a widely used synthetic dataset that simulates a corporation’s activity logs, produced by Carnegie Mellon University. Each version contains several thousand simulated employees observed over about eighteen months, with a small set of employees scripted to carry out malicious scenarios on particular days. Version r4.2 has sixty malicious users; version r6.2 has four. We later add two real datasets: TWOS, collected from real people with real personality questionnaires, and a corporate login log with no profile of any kind; both are described in Section 5.

Each user-day is converted into a short structured text with three kinds of lines:

- a **DAY** line holding static organizational fields for that user (project, role, business unit, department, team, and an information-technology administrator flag);
- a **PSY** line holding the user’s five simulated personality scores. These are the “Big Five” personality traits used in psychology: openness (O), conscientiousness (C), extraversion (E), agreeableness (A), and neuroticism (N). In CERT these numbers are generated by the simulator; they are constant for each user;
- one **SES** line per work session, holding numeric behavioral counts for that session: duration, number of logons, file and removable-drive activity, e-mail volume, web-browsing category counts, and similar quantities.

A user-day text therefore looks like a small table serialized as text. The first two lines describe *who the user is* and are essentially constant across a user’s days; the remaining lines describe *what the user did* that day.

3 The detector

3.1 Language models as probability distributions

A *language model* is a parametric probability distribution over sequences of symbols called tokens. Writing a sequence as $x = (x_1, \dots, x_n)$, the model has the product form

$$p_\theta(x) = \prod_{t=1}^n p_\theta(x_t \mid x_1, \dots, x_{t-1}), \quad (2)$$

where each conditional distribution is computed by a neural network with parameters θ . Modern “large” language models are such networks with billions of parameters, trained on enormous general text corpora. We use one called Qwen3-8B (eight billion parameters). The network processes a sequence through L successive layers; at layer ℓ and position t it maintains an internal state vector $h_t^{(\ell)} \in \mathbb{R}^d$ with $d = 4096$. These internal vectors matter later: they are the objects we will analyze.

3.2 Adapting the model to normal behavior only

We specialize the general-purpose model to our corpus by *fine-tuning*: continuing its training on the serialized user-day texts, using only days from benign users. The training objective is maximum likelihood, that is,

minimizing the negative log-likelihood of the benign corpus $\mathcal{D}$:

$$\mathcal{L}(\theta) = - \sum_{x \in \mathcal{D}} \sum_t \log p_\theta(x_t | x_{<t}). \quad (3)$$

No malicious example is ever used in training. This is the classical one-class setup: model the normal population, then measure deviation from it.

For efficiency the fine-tuning does not modify all eight billion parameters. Each weight matrix W of the network is kept fixed and receives a trainable low-rank correction,

$$W' = W + \frac{\alpha}{r} BA, \quad B \in \mathbb{R}^{d_{\text{out}} \times r}, A \in \mathbb{R}^{r \times d_{\text{in}}}, \quad (4)$$

with rank $r = 16$, so only the small matrices A and B are learned. This technique is called LoRA (low-rank adaptation); we use a memory-saving variant called QLoRA in which the fixed weights are stored at reduced numerical precision. The pair (fixed base model, learned low-rank correction) is called the *adapted model*. The important point is that the entire effect of training is carried by a low-dimensional, removable perturbation, which makes “before versus after” comparisons clean.

3.3 The anomaly score

The score of a day x is its average surprisal under the adapted model:

$$s(x) = -\frac{1}{n} \sum_{t=1}^n \log p_{\theta_{\text{ada}}}(x_t | x_{<t}). \quad (5)$$

Days that the model of normal behavior finds improbable receive high scores. On the standard evaluation protocol for these benchmarks the score ranks very well: AUC about 0.95 on r6.2 and 0.96 on r4.2. By the usual standards of the field, this would be reported as a strong detector. The rest of this document asks what that number is actually made of.

4 Looking inside the model

4.1 What did fine-tuning change?

Because the base model is kept frozen, we can run the same day x through both the base and adapted models and subtract their internal states. Define, at a chosen layer ℓ and each token position t ,

$$\delta_t = h_t^{(\ell), \text{ada}} - h_t^{(\ell), \text{base}} \in \mathbb{R}^d. \quad (6)$$

The vector δ_t isolates the representational change induced by the fine-tuning at the audited layer and position. It is the natural object to study if one wants to know what the adaptation, as opposed to the general-purpose model, contributes at that point in the computation.

4.2 A sparse dictionary for the changes

The vectors δ_t live in $\mathbb{R}^{4096}$ and are not individually interpretable. We therefore approximate them by sparse nonnegative combinations of learned dictionary elements. Concretely, we seek a dictionary matrix $D \in \mathbb{R}^{d \times m}$ (with $m = 4d$ columns, called *features*) and, for each δ_t , a coefficient vector $z_t \in \mathbb{R}_{\geq 0}^m$ with at most $k = 8$ nonzero entries, such that

$$\delta_t \approx Dz_t, \quad (7)$$

with D and the code map chosen to minimize the mean squared reconstruction error over a large sample of token positions. This is a sparse coding problem; in the machine-learning literature the particular parametrization we use (a one-layer encoder network followed by a keep-the-largest- k rule and a linear decoder) is called a *sparse autoencoder*. The outcome is that each change vector is expressed as a short combination of reusable directions, and each direction can be studied on its own.

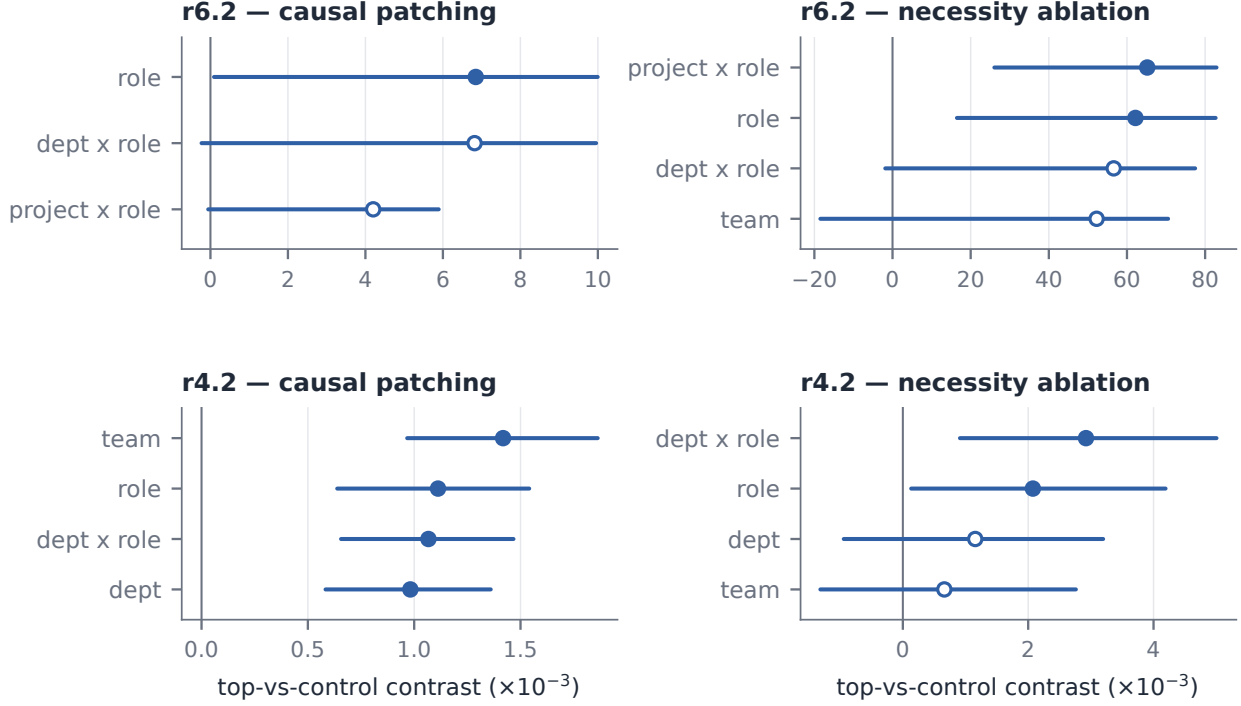


Figure 1: Effect of editing the selected features, compared with editing matched control features, on the anomaly score (one panel per benchmark and per type of edit; units are thousandths of the score). Each point is the average effect for one organizational matching context, and each horizontal bar is a 95 percent confidence interval computed with the malicious users as the resampling unit. Positive values mean that the selected features influence the score more strongly than the control features do. Open circles mark intervals that include zero.

4.3 Selecting candidate features

Some dictionary features activate similarly on malicious and benign days; those cannot carry the anomaly signal. We rank features by the difference in mean activation,

$$g_f = \mathbb{E}[z_{t,f} \mid \text{malicious day}] - \mathbb{E}[z_{t,f} \mid \text{benign day}], \quad (8)$$

and take the five features with the largest g_f as the candidate mechanism. The full paper verifies, through intervention experiments, that these selected features genuinely influence the anomaly score: if one edits the coefficients $z_{t,f}$ of the selected features and propagates the edited internal state through the rest of the network, the score changes, and it changes more than when the same edit is applied to matched control features. We omit the details of those experiments here and take away only their conclusion: the selected features are causally connected to the score, not merely correlated with it. Figure 1 summarizes those measurements.

4.4 Reading off what the features respond to

Finally we ask *where* the selected features activate within the day text. Every token position belongs to one line class: the organizational **DAY** line, the personality **PSY** line, or a behavioral **SES** line. For a feature f , define its *attribution mass* on class c as the fraction of its total activation that occurs at positions of that class:

$$m_f(c) = \frac{\sum_{(x,t) : \text{class}(t)=c} z_{t,f}(x)}{\sum_{(x,t)} z_{t,f}(x)}. \quad (9)$$

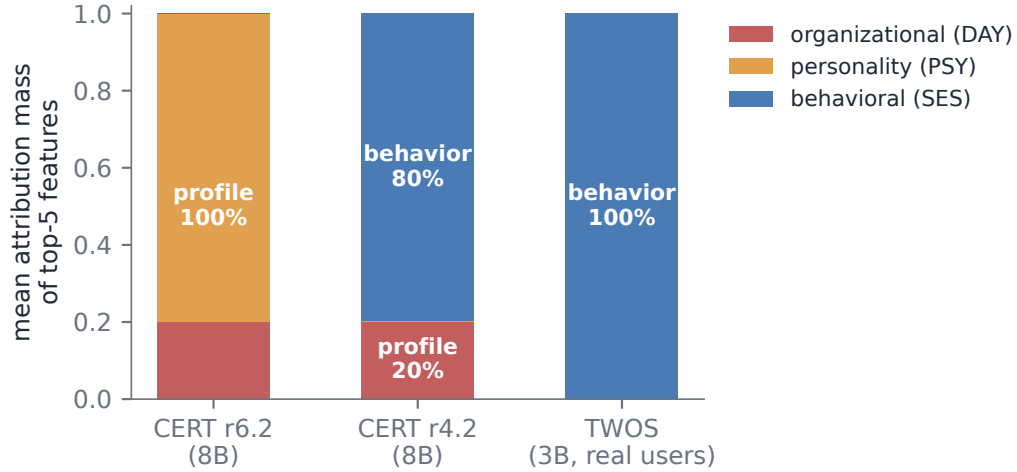


Figure 2: Where the selected features place their activation, averaged over each detector’s five selected features. Each bar decomposes the activation mass into the organizational lines, the personality lines, and the behavioral session lines of the input text. The r6.2 detector’s selected features respond almost entirely to who the user is; the r4.2 and TWOS detectors’ selected features respond almost entirely to what the user did.

If a feature has nearly all of its mass on PSY and DAY positions, it responds to who the user is; if its mass lies on SES positions, it responds to what the user did. This single quantity is what turns the analysis into an answer to our opening question.

5 Findings

5.1 Two detectors that look alike and are not

The two CERT versions produce nearly identical headline AUC values, yet the analysis above reveals qualitatively different detectors.

On r6.2, the five selected features place between 99.8 and 100 percent of their attribution mass on the personality and organizational lines. The evidence the audit selects, the five features that most raise the score, is almost entirely a description of *who the user is*. On r4.2, four of the five selected features concentrate on the behavioral session lines. There, the evidence is mostly *what the user did*. Figure 2 summarizes this contrast, together with the real-data result of Section 5.3.

5.2 The consequence for the headline numbers

Why would reading a user’s static profile produce a high AUC at all? The adapted model was trained on roughly ninety percent of the benign users. For those users, the model has been repeatedly exposed to the constant profile lines, so their profile text is comparatively familiar and their days look unsurprising. The malicious users, however, were never seen in training, so their profile lines alone make their days look surprising. The score is then partly measuring *familiarity* rather than behavior.

This suggests a sharper test: compare malicious days only against benign days of users who were also excluded from training, so that both sides are equally unfamiliar. Under this comparison the AUC falls from 0.95 to 0.55 on r6.2, which is barely better than chance, and from 0.96 to 0.67 on r4.2. The apparently excellent detector on r6.2 was, to a first approximation, an unfamiliar-user detector. Consistently with this, deleting the personality lines from the inputs at scoring time *improves* the r4.2 detector’s performance on

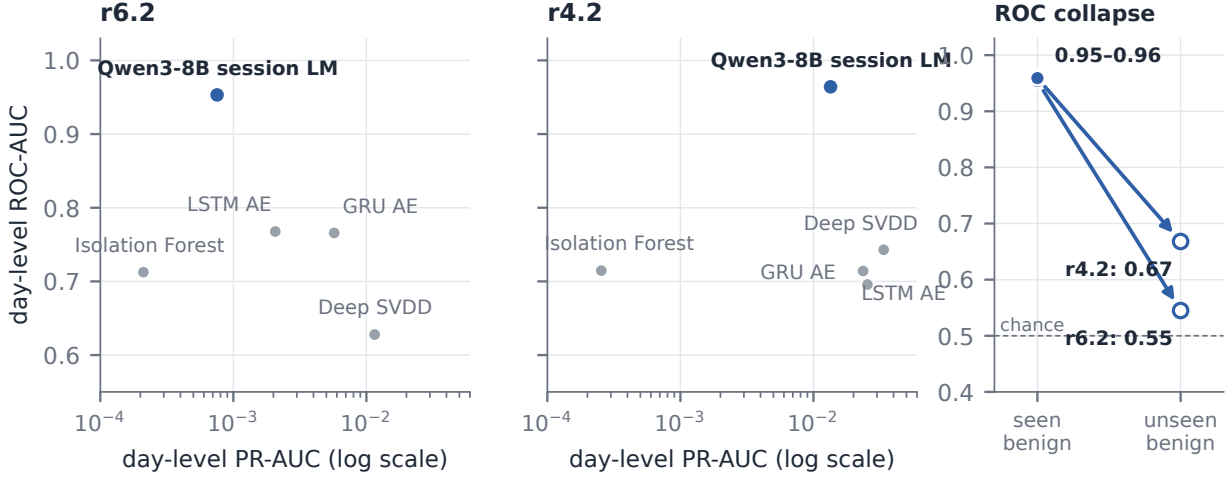


Figure 3: Score-level results. Left and center panels: ranking quality (vertical axis) and average precision, the area under the precision–recall curve (horizontal axis, logarithmic scale) for the language-model detector (blue) and four standard anomaly-detection methods on the two benchmarks; the language model attains the highest ranking quality under the standard protocol. Right panel: the same language-model ranking quality when malicious users are compared only against benign users never seen during training. The apparent advantage largely disappears (from 0.95 to 0.55 on r6.2 and from 0.96 to 0.67 on r4.2), and the dashed line marks chance level.

unseen users from 0.67 to about 0.80: the profile text was not merely unhelpful, it was actively obscuring the behavioral signal. Figure 3 shows the comparison with standard detectors and the collapse.

5.3 A real-data test with real personality scores

The CERT data are simulated, including the personality scores, so we repeated the entire pipeline on TWOS, a dataset collected from twenty-four real users over one week, during which some users, by design of the underlying study, temporarily took over other users’ sessions. Each real user completed a standard fifty-item Big Five personality questionnaire, so the serialized days contain genuine psychometric information. In this dataset half of the users experience an attack, so profile information is a far weaker separator than in CERT r6.2, and our account predicts that the detector should rely on behavior. That is what the analysis finds: the selected features place essentially all of their mass on the behavioral lines, under two independent training runs. Decisively, the evaluation compares each user’s attacked windows with that same user’s normal windows, so static identity and personality information is held fixed and cannot explain the observed separation (average within-user AUC 0.78 over the 16 users with attacked windows).

5.4 A second real-data test, with no profile at all

TWOS still contains a profile line, albeit a real one. To remove profile information entirely we turned to a corporate login log released by Los Alamos National Laboratory in 2015: fifty-eight days of authentication events from a real enterprise (12,425 users, about a billion events), during which a security “red team” carried out 749 documented intrusions using stolen credentials. Each event records only who logged on, from which computer, to which computer, by what mechanism, and whether it succeeded; there is no personality line and no organizational header. We grouped each user’s events into windows of thirty-two consecutive events, called a window malicious if it contained a red-team event, adapted the smaller three-billion-parameter sibling of our model to benign windows only, and scored held-out windows exactly as before. Users were divided into five groups; the model was adapted on four and the fifth was held out, so ranking quality can be reported separately for *seen* users (in the adaptation groups) and *unseen* users (the held-out group).

Because identity now sits in two fields, we trained four versions of the detector on four versions of the text: the full line; the user name hidden; the computer names hidden; and the user names scrambled, each user being given another user’s name throughout.

Identity fields in the text	Seen-user AUC	Unseen-user AUC
full	0.949	0.497
user name hidden	0.944	0.518
computer names hidden	0.877	0.627
user names scrambled	0.947	0.488

The pattern of the CERT experiments returns in a dataset that has no profile text at all. Every version ranks seen users well and unseen users far worse: at chance, except when the computer names are hidden. Hiding or scrambling the user name changes little (at most 0.02 on unseen users; scrambling with a fixed permutation relabels a user rather than removing the label), so the shortcut is not mainly the user name. Hiding the computer names costs some seen-user quality and buys unseen-user quality, the trade-off produced by CERT’s profile lines, although the fraction of alerts that would be correct on unseen users stays at the base rate in every version (average precision about 0.013 at a prevalence of 0.65 percent), so the gain is in ranking, not in usable alerts. Computer names are associated with users without being fixed per user: a typical user’s most frequent computer accounts for about half of that user’s events (median 54 percent), and only 7 percent of users issue at least 80 percent of their events from one computer. The computer name is therefore a partial proxy for identity that may also encode which resources a user reaches, so its analogy with CERT’s fixed personality line is qualitative. A gap between seen and unseen users remains after the computer names are hidden (0.88 versus 0.63), which may reflect genuine differences between people’s behavior, residual identity cues, or both. Two cautions apply: each of these four detectors was trained once, and the intrusions are staged credential theft rather than the insider scenarios of CERT, so the numbers replicate the mechanism rather than benchmark a product.

5.5 Retraining without the profile: is the profile the cause?

Everything so far shows that a detector trained *with* the profile present relies on it. A skeptic can still ask whether the profile is the cause of the failure on unseen users or merely something the detector happens to notice. The direct test is a controlled experiment: train several detectors that are identical in every respect except the profile block, and compare them on the same unseen users. We trained four such detectors on CERT r6.2, at both the eight-billion-parameter model of the main paper and its three-billion-parameter sibling: (i) the full text; (ii) the personality line removed; (iii) both profile lines removed; (iv) the profiles scrambled, meaning that every user is given another user’s profile throughout, so the text keeps the same form, the same statistics, and the same constancy within each user, but the profile no longer describes its owner.

For each detector we used the strictest evaluation available. Each of the four malicious users in r6.2 is held out in turn and ranked against all 406 benign users of this pool that were never used in training (the protocol’s cap of 800 exceeds the pool); a user’s score is the largest of its daily scores; the AUC is averaged over the four held-out users. Uncertainty comes from resampling the malicious users ten thousand times and recomputing everything, which yields an interval for each number and, because the same users are resampled for every detector, a paired interval for the difference between two detectors. With only four malicious users these intervals are descriptive: they say how much the average depends on which users were held out, for these particular trained detectors, and nothing about how a retrained detector would differ.

Text the detector was trained on	3B model	8B model
full	0.926	0.532
personality line removed	0.903	0.906
both profile lines removed	0.956	0.903
profiles scrambled	0.656	0.485
removal effect (both removed – full)	+0.03 [−0.07, +0.14]	+0.37 [+0.07, +0.69]

Read the 8B column first, since that is the model of the main paper. With the profile present, the detector ranks unseen users at chance (0.532). With the profile removed it ranks them at 0.903. The difference, +0.37, has a resampling interval that excludes zero: for this detector the profile is not merely noticed; its presence in the text used for training and scoring *causes* the failure on unseen users, and removing it repairs the detector. (Three of the four users improve, by +0.29 to +0.82; the fourth, already caught, slips by 0.05.) Removing the personality line alone repairs it just as well (0.906), so that line is the carrier.

Now read the 3B column. The smaller model does not fail in the first place: with the profile present it already ranks unseen users at 0.926, and removing the profile changes little (+0.03, an interval that includes zero). Yet when we open the small model by the method of Section 4, its selected features are just as profile-bound as the large model’s. The small model has *noticed* the profile without being measurably harmed by it. It is tempting to read this as a size effect, and it is consistent with the capacity explanation offered later, but the two models differ in more than size (they come from different model generations with different pre-training), each was trained once, and a difference that is not statistically detectable is not a demonstrated absence. What the experiment establishes is the causal role of the profile in the large model we audited.

Looking user by user makes the mechanism concrete. The large model’s failure is not spread evenly: one of the four malicious users is still caught, two are ranked at chance, and one is ranked among the *least* suspicious of the 407 users compared, worse than a coin flip. That last user is the one whose five personality scores most closely resemble those of many benign training users (12 such users nearby), and the user still caught has few such neighbours (3); the ordering is only approximate, since the user with no close neighbours at all is one of the two ranked at chance. A large model trained on normal users learns that familiar personality scores mean “normal,” and a malicious user who happens to have ordinary scores can be waved through. With the profile removed, all four users are ranked in the top fifth of the 407 (positions 10 to 75 for the large model, 4 to 64 for the small one); the user who was already caught slips slightly, the other three rise. Scrambling the profiles never helps: a fixed description that belongs to someone else is at best useless and at worst misleading, even though it is just as constant within each user as the true one. One further observation deserves recording. When we open the scrambled-profile model, its selected features are still profile-bound, exactly as in the honest-profile model, even though its ranking of unseen users has collapsed. Where a model *looks* and what *helps* it are different things, which is why this paper tests features by intervening on them rather than by attribution alone.

5.6 Where the shortcut comes from

One might suspect that the r6.2 data simply force any statistical method to use profile information. This is testable. We trained three classical anomaly detectors (an isolation forest, a one-class support vector machine, and a principal-component reconstruction method; their details do not matter here) on the same days, represented as ordinary numeric feature vectors that *include* the profile variables. All three essentially ignore the profile variables: those variables receive about four to seven percent of the total importance, no profile variable appears among any method’s top five, and deleting the profile variables does not hurt their performance.

The shortcut is therefore not an inevitable consequence of the available profile information: we do not observe comparable profile reliance in the three classical detectors. An explanation consistent with this evidence lies in the training objective. For a likelihood-based sequence model, the per-user-constant profile tokens are the most predictable part of every training sequence, so learning them yields a large and cheap reduction of the training loss. In a plain feature-vector representation, the same information is twelve columns among two hundred forty-two, with no comparable incentive attached. Our summary of the whole phenomenon

is a two-factor statement: the population structure of a dataset determines *when* an identity shortcut is available, and the training objective together with the data representation determines *which* detectors take it. The retraining experiment adds that, in the large model we audited, the profile’s presence in the text causes the failure rather than merely accompanying it, and the login-log experiment adds that the shortcut lands on whichever field reveals identity, even a field that is only associated with a user rather than fixed: the computer name there, the personality line here.

6 Why this matters

Three conclusions carry beyond these particular datasets and this particular model.

First, for anomaly detectors intended to detect behavior, ranking quality alone is not evidence of success. A detector should be evaluated with the unfamiliarity confound removed, by comparing malicious observations only against normal observations from equally unseen users, and, where possible, by within-user comparisons.

Second, including static descriptors of people (role, department, personality measures) in a detector’s input is not harmless. In the likelihood-based detector it created the shortcut, and even where the detector was behavioral it degraded performance. Retraining the large model with the profile removed repairs its ranking of unseen users from chance to about 0.90.

Third, the question “what does this detector actually detect?” can be answered, not merely asked. The chain presented here, namely differences of internal states, a sparse dictionary approximation, selection of score-relevant directions, and attribution of those directions to input content, turns the question into a measurable quantity. In the full paper this chain is supported by causal intervention experiments and an extensive robustness program, and it is that combination, rather than any single number, that justifies the conclusions summarized here.